Finite-tree automorphism actions are Tame_1 but not Tame_2

A candidate partial answer to Question 5.12 of arXiv:2601.01681

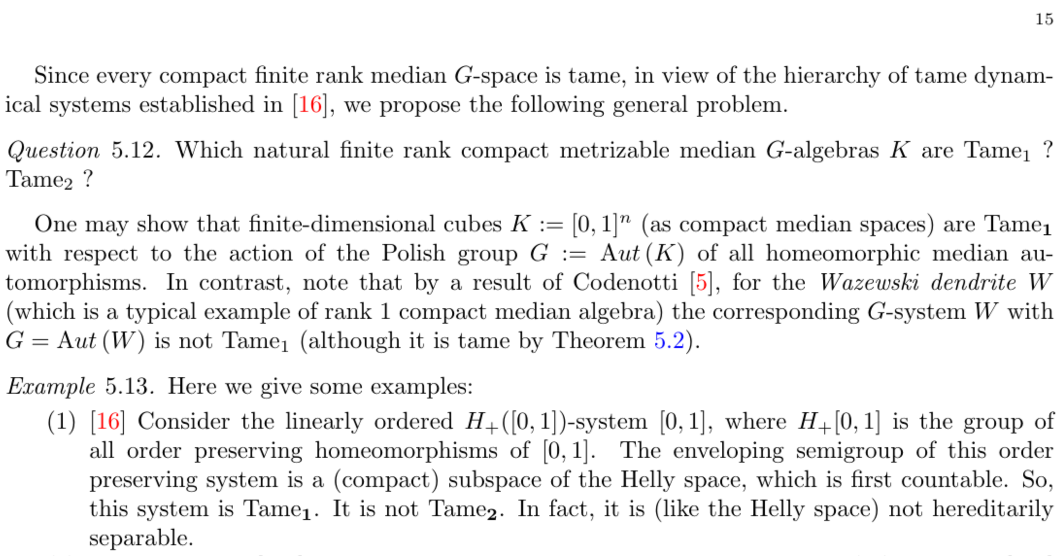
Automated mathematics run `fa_banach_001`, lane 3

9 August 2026

Status. Candidate partial result; likely valid; awaiting human review.

1 Source question and scope

Megrelishvili asks which natural finite-rank compact metrizable median G-algebras are Tame_1 or Tame_2 [1, Question 5.12]. The source gives the interval and finite-dimensional cubes as Tame_1 examples, and contrasts them with the Ważewski dendrite, whose full automorphism action is not Tame_1.



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Since every compact finite rank median G-space is tame, in view of the hierarchy of tame dynamical systems established in [16], we propose the following general problem.

Question 5.12. Which natural finite rank compact metrizable median G-algebras K are Tame_1 ? Tame_2 ?

One may show that finite-dimensional cubes $K := [0,1]^n$ (as compact median spaces) are $\mathrm{Tame}_{\mathbf{1}}$ with respect to the action of the Polish group $G := \mathrm{A}ut\,(K)$ of all homeomorphic median automorphisms. In contrast, note that by a result of Codenotti [5], for the *Wazewski dendrite* W (which is a typical example of rank 1 compact median algebra) the corresponding G-system W with $G = \mathrm{A}ut\,(W)$ is not Tame_1 (although it is tame by Theorem 5.2).

Example 5.13. Here we give some examples:

(1) [16] Consider the linearly ordered $H_+([0,1])$-system $[0,1]$, where $H_+[0,1]$ is the group of all order preserving homeomorphisms of $[0,1]$. The enveloping semigroup of this order preserving system is a (compact) subspace of the Helly space, which is first countable. So, this system is $\mathrm{Tame}_{\mathbf{1}}$. It is not $\mathrm{Tame}_{\mathbf{2}}$. In fact, it is (like the Helly space) not hereditarily separable.

(2) Consider $K := [0,1]$ as a linear compact median algebra and $G = \mathrm{A}ut\,(K) = \mathrm{Homeo}\,[0,1]$

Transcription. "Which natural finite rank compact metrizable median G-algebras K are Tame_1? Tame_2?"

This packet gives a complete classification for the natural rank-one class of geometric realizations of finite simplicial trees. It does not answer the source question for arbitrary finite-rank median algebras or arbitrary group actions.

2 Definitions

Let K be the geometric realization of a finite simplicial tree. Its median $m(x,y,z)$ is the unique point in

$$[x,y] \cap [y,z] \cap [z,x],$$

where $[x, y]$ denotes the unique arc from x to y. We give $G = \mathrm{Aut}(K)$ the compact-open topology and let it act naturally on K.

For a compact G-space K, its enveloping (Ellis) semigroup is

$$E(G, K) = \overline{\{g : K \to K : g \in G\}}^{K^K},$$

where the closure is pointwise. In the terminology used by the source, (G, K) is Tame$_1$ when $E(G, K)$ is first countable, and Tame$_2$ when $E(G, K)$ is hereditarily separable (within the tame class).

The source recalls that the enveloping semigroup of the order-preserving homeomorphism action on $[0, 1]$ is a compact subspace of the first-countable Helly space of monotone maps, but is not hereditarily separable [1, Example 5.13]; see also [2].

3 Proof intuition

A finite tree has a finite topological skeleton: its endpoints and branch points. Removing this skeleton leaves finitely many open edges. Every homeomorphism permutes the skeleton and the edges. After fixing this finite permutation, an automorphism is described on each edge by a monotone interval homeomorphism. Pointwise limits are therefore controlled by finitely many Helly compacta, which gives first countability.

The negative Tame$_2$ conclusion comes from looking in the opposite direction: one edge already carries all order-preserving interval homeomorphisms, extended by the identity on the rest of the tree. Their non-hereditarily-separable Ellis semigroup embeds in the Ellis semigroup of the full tree action.

4 The partial result

Theorem 1. *Let K be the geometric realization of a nondegenerate finite simplicial tree, equipped with its canonical median, and let $G = \mathrm{Aut}(K)$ be the compact-open group of all homeomorphic median automorphisms. Then the natural compact median G-algebra (G, K) is Tame$_1$ but not Tame$_2$.*

Proof. Every homeomorphism of a tree preserves unique arcs and therefore preserves their triple-intersection median. Thus $G = \mathrm{Homeo}(K)$ as sets.

Let V be the set of endpoints and branch points of K. It is finite and is preserved by every homeomorphism. The connected components of $K \setminus V$ are finitely many open arcs; their closures will be called the essential edges. Every $g \in G$ permutes V and the essential edges.

Put $E = E(G, K)$. If $p \in E$, choose a net (g_i) in G converging pointwise to p. For each $v \in V$, the net $(g_i v)$ takes values in the finite set V. Its convergence forces it to be eventually constant. Since V is finite, on one common tail all $g_i|_V$ equal a single permutation σ. Consequently every $p \in E$ belongs to

$$E_\sigma := \overline{\{g \in G : g|_V = \sigma\}}^{K^K}$$

for one of finitely many realizable permutations σ.

These pieces are pairwise disjoint and clopen in E: evaluation on the finite invariant set V separates the possible restrictions. Fix σ. It also fixes the target essential edge $\sigma(e)$ of each essential edge e. Choose coordinates $\theta_e : [0,1] \to e$. After reversing one coordinate when the endpoint order demands it, every normalized restriction

$$\theta_{\sigma(e)}^{-1} \circ g|_e \circ \theta_e$$

is an increasing homeomorphism of $[0,1]$. Hence restriction to the finitely many edges defines a continuous injection of E_σ into a finite product of Helly compacta of nondecreasing interval maps. Since E_σ is compact and the product is Hausdorff, this injection is a topological embedding. The Helly compactum is first countable [1, Example 5.13]; so finite products and their subspaces are first countable. Thus every E_σ is first countable. Because the finitely many E_σ are clopen, E is first countable. Therefore (G, K) is Tame_1.

It remains to rule out Tame_2. Fix one essential edge e and identify it with $[0,1]$. Every increasing homeomorphism of e fixing its endpoints extends by the identity on $K \setminus e$ to an element of G. Let H be this subgroup. Extension by the identity induces a topological embedding

$$E(H_+([0,1]), [0,1]) \hookrightarrow E(H, K) \subseteq E(G, K).$$

The interval enveloping semigroup on the left is not hereditarily separable [1, Example 5.13]. Hence $E(G, K)$ is not hereditarily separable. The system is not Tame_2. □

5 Verification notes

The proof deliberately uses the fact that the skeleton-permutation pieces are *clopen*. An arbitrary finite union of closed first-countable subspaces need not be first countable, so merely writing the Ellis semigroup as such a union would not suffice. Evaluation on the finite invariant skeleton supplies the needed clopen decomposition.

The interval case is recovered when V consists of two endpoints and there is one essential edge. Adding or deleting valence-two subdivision vertices does not change V, the essential edges, or the argument. A triod is the first branched example: the center and three endpoints give a four-point skeleton and three Helly coordinates in each permutation component.

No computational step is used. The proof depends externally only on the first-countability and non-hereditary-separability facts for the classical interval Ellis semigroup, both explicitly recorded in the source's Example 5.13.

6 Limitations and novelty check

This is a partial answer: it treats the full automorphism action on finite topological trees, not arbitrary actions on dendrites and not higher-rank median spaces. It also does not classify which infinite dendrites have Tame_1 full automorphism actions.

On 9 August 2026, the run's lightweight registry, solution, attempt, and proof-gap indexes were searched for arXiv:2601.01681 and finite-rank median/tameness terms, with no hit. A bounded web/arXiv search used combinations of "finite tree", "automorphism", "Tame_1", "dendrite", and "enveloping semigroup". It found the source paper and Codenotti's dendrite counterexamples [3], but no prior finite-tree classification. Novelty confidence is therefore moderate, not definitive.

7 Human-review recommendation

Send to a specialist in tame dynamics or continuum theory. The highest-value checks are: (i) that the essential-skeleton restriction indeed yields clopen Ellis components, and (ii) that extension by the identity embeds the complete interval Ellis semigroup, not merely the acting group.

References

[1] M. Megrelishvili, *Tameness of actions on finite rank median algebras*, arXiv:2601.01681, April 2026.

[2] E. Glasner and M. Megrelishvili, *Todorčević's Trichotomy and a hierarchy in the class of tame dynamical systems*, Trans. Amer. Math. Soc. **375** (2022), 4513–4548.

[3] A. Codenotti, *Some examples of tame dynamical systems answering questions of Glasner and Megrelishvili*, Proc. Amer. Math. Soc. **153** (2025), 2433–2449; arXiv:2305.00726.